

agent-smith

An evolving agent system build on langchain/langgraph

“The Matrix cannot tell you who you are.”

agent-smith is a modular, extensible framework for building agentic workflows and autonomous agents across diverse functional domains. It provides the infrastructure, orchestration primitives, and a growing library of pre-built agents — so you can focus on what your agents *do*, not on how to wire them together. Take care. We haven’t took the red pill yet! It’s just the beginning.

Keep in mind that this code is for learning, experimenting and demonstrating purpose only! Don’t use it for production scenarios. Never use it for use cases where security is an issue! —

What is agent-smith?

agent-smith enables you to:

- **Compose complex workflows** from simple, reusable agent building blocks
- **Deploy specialized agents** for tasks like data retrieval, code execution, document processing, web search, API interaction, and more
- **Orchestrate multi-agent pipelines** where agents collaborate, delegate, and hand off work to each other
- **Integrate with any LLM backend** — bring your own model or use the defaults

Whether you need a single autonomous agent or a network of cooperating agents tackling a multi-step problem, Agent Smith gives you the scaffolding to build it.

Features

- **Modular agent architecture** — every agent is self-contained and composable
- **Workflow orchestration** — define sequential, parallel, or conditional agent pipelines
- **Tool use & function calling** — agents can use tools, call APIs, and interact with external systems
- **Memory & context management** — short-term and long-term memory abstractions out of the box
- **Extensible agent library** — add your own agents or use the built-in ones
- **LLM-agnostic** — works with OpenAI, Anthropic, local models, and any OpenAI-compatible API

- **Observability** — built-in logging and tracing for every agent step

Built-in Agents

Agent	Description
WebSearchAgent	Searches the web and summarizes results
CodeExecutionAgent	Writes and executes code in a sandboxed environment
DocumentAgent	Reads, summarizes, and extracts information from documents
APIAgent	Calls REST APIs and processes responses
DataAgent	Queries, transforms, and analyzes structured data
OrchestratorAgent	Coordinates other agents to solve complex multi-step tasks

Dependencies

```
pip install -r requirements.txt
pip install --upgrade setuptools
pip install -e .
```

You need a model that supports tool use! The default model used is *Mistral:latest* with ollama. You can easily pull it with

```
ollama pull mistral:latest
```

Quickstart

```
pip install agent-smith

from agent_smith import AgentSmith, WebSearchAgent, OrchestratorAgent

smith = AgentSmith()

# Register agents
smith.register(WebSearchAgent())
smith.register(OrchestratorAgent())

# Run a workflow
```

```
result = smith.run("Research the latest developments in optical interferometry and write a s  
print(result)
```

Architecture

Hier ist das Diagramm der AgentSmith Core-Komponenten:

```
flowchart TD
    subgraph AgentSmith Core
        direction TB
        Orchestrator --> AgentRuntime
        ToolRegistry --> AgentRuntime
        AgentRuntime --> Memory
        AgentRuntime --> LLM_Layer
    end

    style AgentSmith Core fill\:none,stroke\:none
```

Contributing

Contributions are welcome! If you want to add a new agent, improve the orchestration engine, or fix a bug:

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/my-new-agent`)
3. Commit your changes
4. Open a pull request

Please make sure your code is well-documented and includes tests.

Run tests

““

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